CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE NAME			
CLASS	2T	INDEX NUMBER	

BIOLOGY
8876/02

Paper 2 Structured and Free-response Questions

24 AUGUST 2021

2 hours

Candidates answer on Question Paper.
No Additional Material are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1 [12]	
2 [5]	
3 [11]	
4 [10]	
5 [7]	
Section B	
6 [15]	
7 [15]	
TOTAL	60

Answer **all** questions.

1 Fig. 1.1 shows an electron micrograph of a cell.

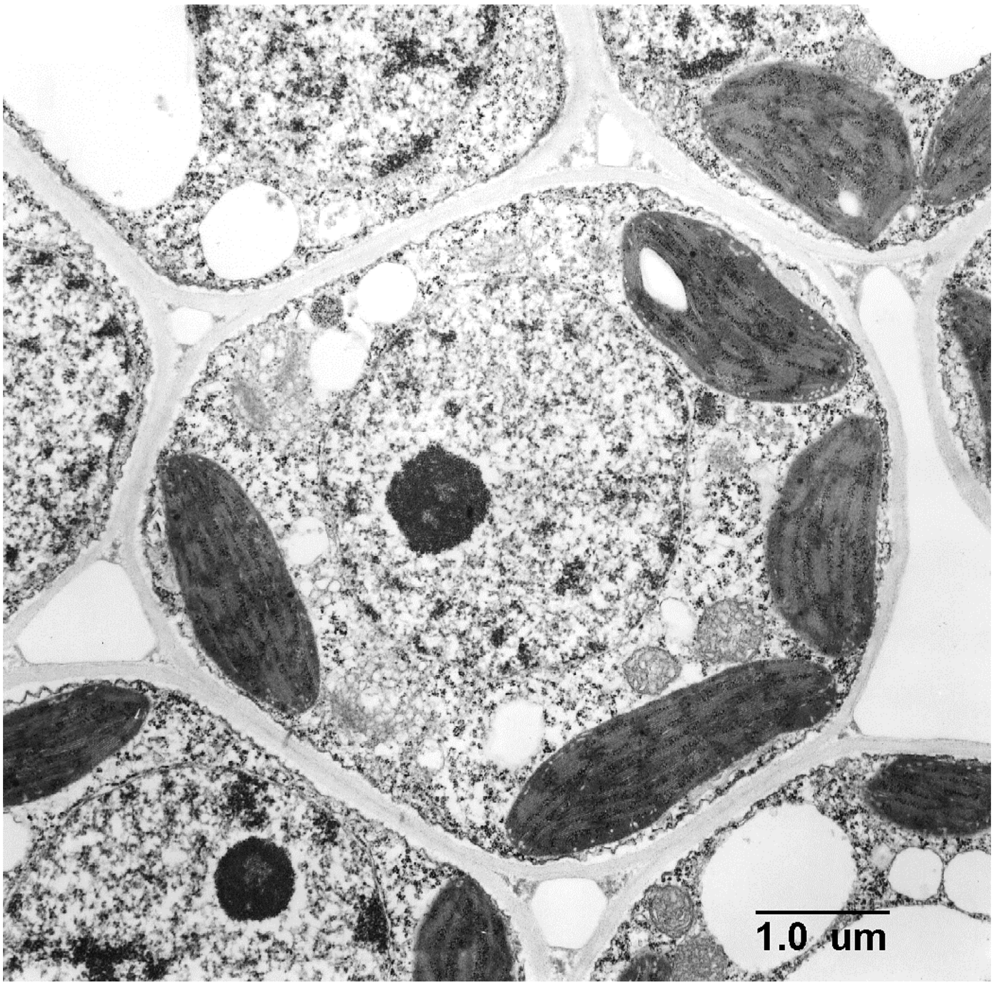

Fig. 1.1

(a) With reference to the Fig 1.1, state two reasons why the cell is considered to be a eukaryotic cell.

Use 2 separate label lines, labelled **1** and **2**, to clearly mark out features visible in Fig. 1.1 to support your answer.

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2. ……….……………………………………………………………………..………………..

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Stachyose is a storage oligosaccharide found in beans, peas and other legumes.

Stachyose can only be metabolized by anaerobic microorganisms in the large intestine. It is responsible for the production of carbon dioxide, hydrogen sulfide and other gases which collectively make up what is known as flatulence.

The structure of stachyose is shown in Fig. 1.2.

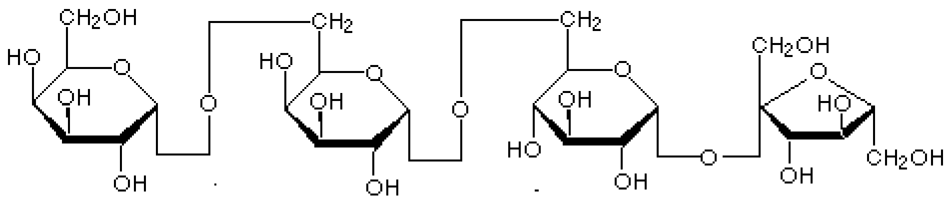

Fig. 1.2

(b) Define the term *glycosidic bond*.

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(c) Several commercial products are available to assist in the digestion of oligosaccharides like stachyose. Suggest how such products prevent the production of flatulence.

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(d) When small amounts of paper containing cellulose are ingested by children, there is usually no gas production. Suggest one reason and a possible consequence of such an event.

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An enzyme known as α-galactosidase cleaves α-(1,6) glycosidic bonds to sequentially release terminal galactose residues. The breakdown of stachyose by α-galactosidase is a two-step process. In the first step of this process, stachyose is hydrolysed to galactose and raffinose.

(e) With reference to Fig. 1.2, draw the products of the <u>first</u> step in this hydrolysis.

[2]

Fig. 1.3 illustrates the activity of α-galactosidase on the substrate stachyose over a period of time.

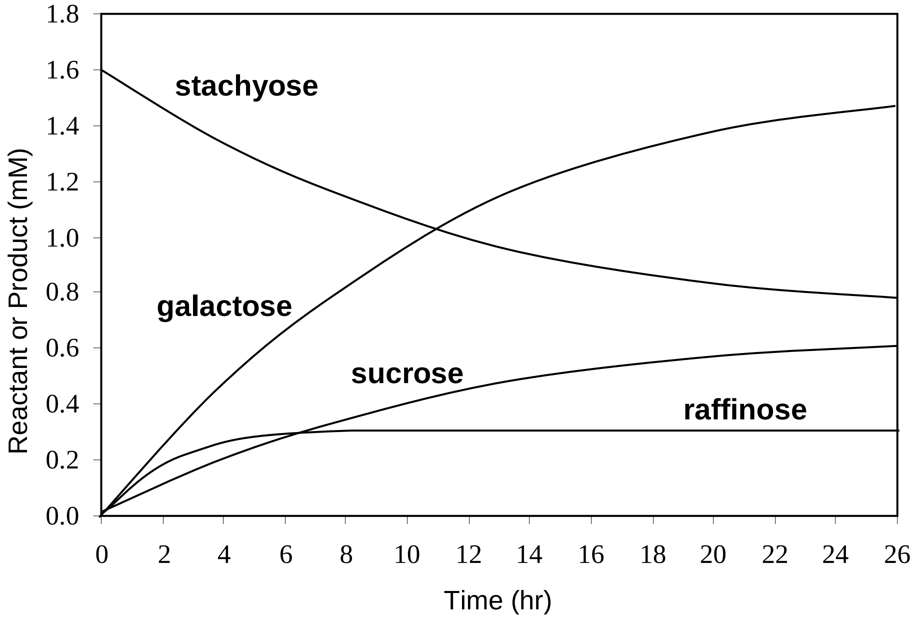

Fig. 1.3

(f) With reference to Fig. 1.3, explain why the level of raffinose remains relatively constant.

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(g) Suggest why the rate at which stachyose is broken down decreases as the concentration of galactose increases.

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[Total: 12]

2 **(a)** The “trombone” model of DNA replication postulates that two DNA polymerase enzymes work together in a protein complex.

In Fig. 2.1, structures labelled **P, Q** and **R** are enzymes involved in the process of DNA replication. Connecting proteins join the two DNA polymerase molecules to enzyme **P**, forming the protein complex.

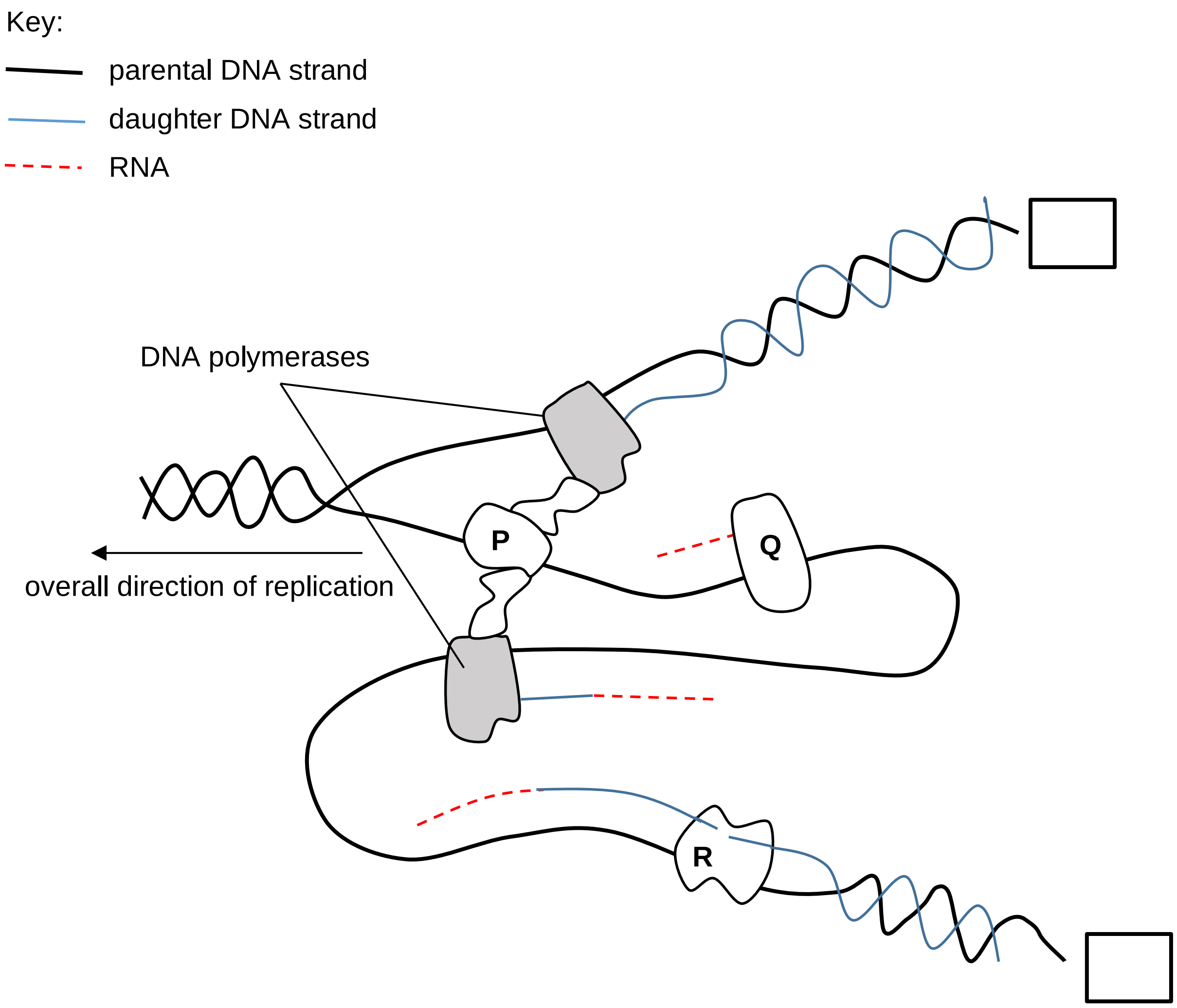

Fig. 2.1

(i) Complete the boxes to indicate clearly the 5’ and 3’ ends of the parental strands shown in Fig. 2.1. [1]

(ii) Identify enzymes **P, Q** and **R**. [2]

P: ……………………………………………………………………

Q: ……………………………………………………………………

R: ……………………………………………………………………

Fig 2.2 illustrates the binding of a protein to a certain region of a DNA molecule.

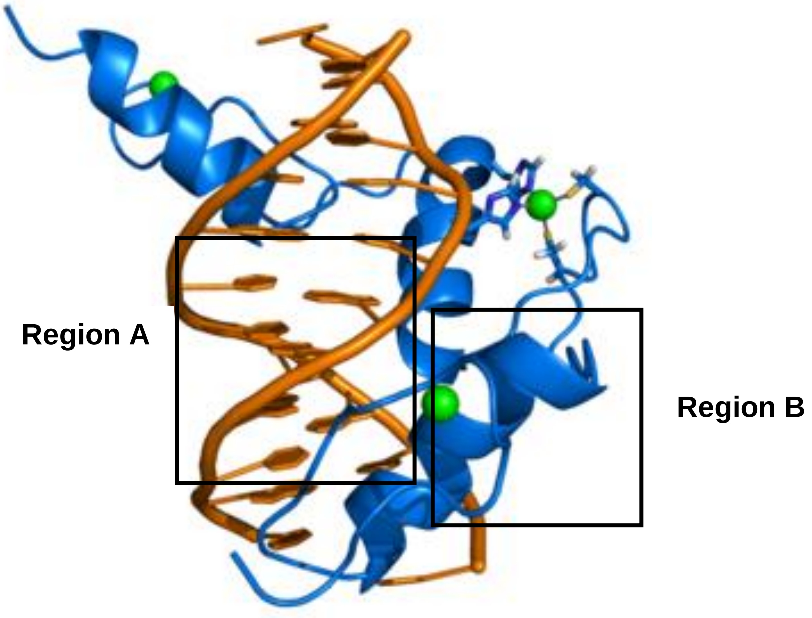

Fig 2.2

(b) With reference to Fig. 2.2, describe how the helical structure shown in **Region A** differs from that of **Region B**.

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[Total: 5]

3 Fig. 3.1 shows the structure of molecules and some associated processes found in a eukaryotic cell.

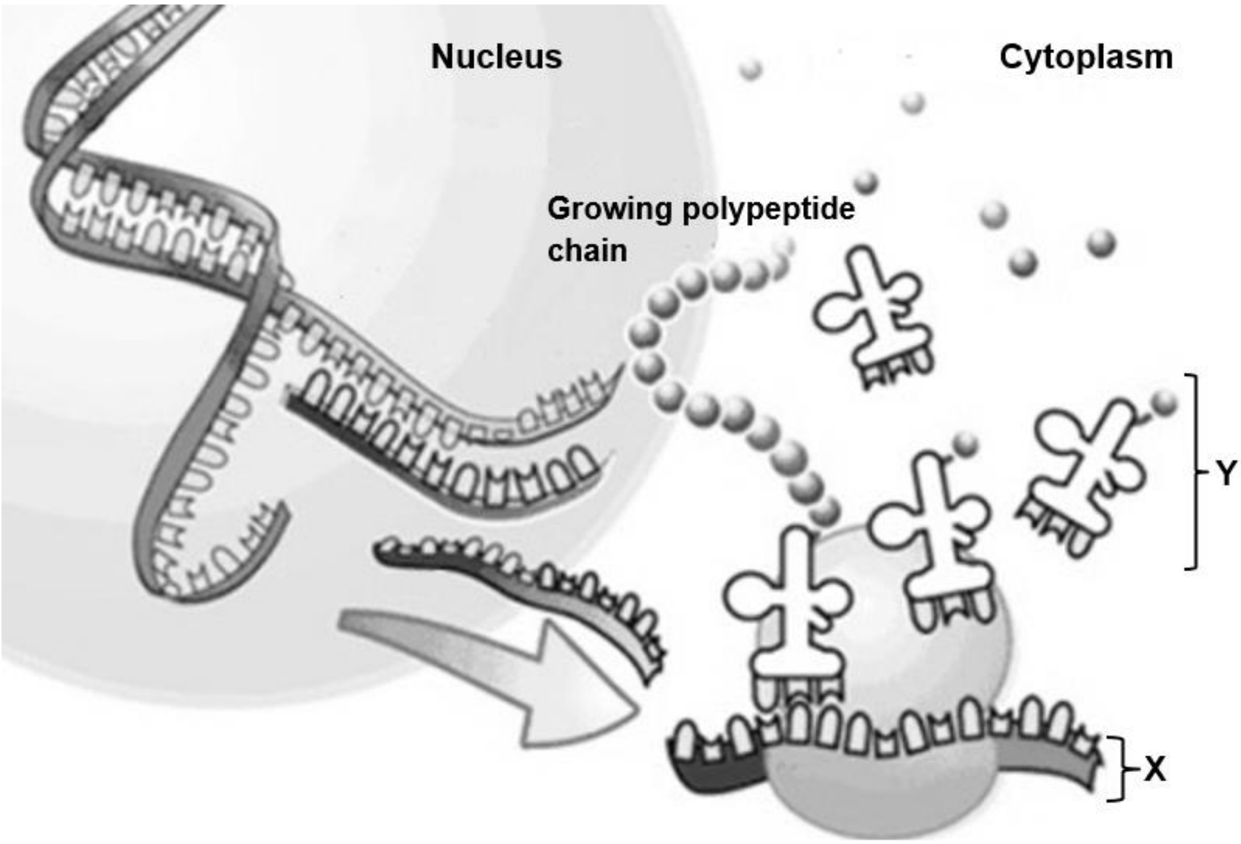

Fig. 3.1

(a) With reference to Fig. 3.1, outline the events that has occurred in the nucleus that synthesised structure **X**.

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…… [4]

(b) Identify structure **Y**.

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(c) Parasites cause diverse types of disease, for example, malaria.

Drug treatments are required to address varying causes of pathogenesis. A new class of drugs has been deployed to block the formation of structure **Y** in parasites.

Suggest how the use of such drugs is effective in treating parasites infections.

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Epidermal growth factor (EGF) is a protein composed of 53 amino acids. It binds to specific cell surface membrane receptors to promote cell proliferation. An example of such surface membrane receptors is the HERS receptor which is a glycoprotein found on the cell surface membrane.

(d) **(i)** Based on the information above**,** explain why EGF cannot move across cell surface membrane through channel proteins.

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(ii) Explain how the structure of HER2 receptor allows it to be embedded across cell surface membrane and serve as a transmembrane protein.

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[Total: 11]

4 About one third of the leg injuries to racehorses involve tendon damage.

In 2006, bone marrow stem cells were taken from injured racehorses and cultured so that they divided many times by mitosis. Each horse's cells (usually around 10 million cells for one tendon) were then injected into its damaged tendons.

80% of the treated horses returned to racing, compared with 30% of those treated conventionally.

(a) **(i)** Stem cells provide the means of maintenance and repair of tissue. With reference to the mitotic cell cycle, explain how the initiation of cell division is controlled.

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(ii) Explain how it is possible that bone marrow stem cells could differentiate into the range of cell types needed for repairing the tendon injuries.

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(iii) Suggest an advantage of the above stem cell therapy.

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(iv) Similar to stem cells, cancer cells are also capable to dividing many times via mitosis.

State 2 differences between stem cells and cancer cells.

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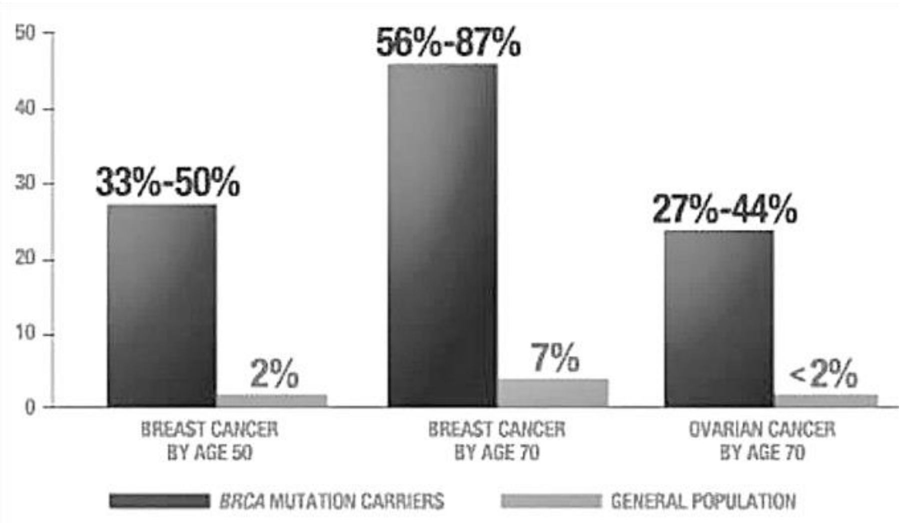
Hereditary breast and ovarian cancer (HBOC) syndrome is an inherited condition that causes an increased risk of developing these cancers, often before the age of 50.

Research shows that about 10% of breast and ovarian cancer is due to an alteration in either of 2 genes: BRCA1 and BRCA2. These genes can come from either your mother or your father.

BRCA1 is a human tumour suppressor gene (also known as a caretaker gene) and is responsible for repairing DNA. BRCA1 and BRCA2 are unrelated proteins, but both are normally expressed in the cells of breast and other tissue, where they help repair damaged DNA, or destroy cells if DNA cannot be repaired.

The figure below shows data concerning breast cancer as a result of mutations in the BRCA genes 1 & 2 on chromosome 17 and 13 respectively. Ovarian cancer data shows a link to BRCA cancer especially by age 70.



(b) Explain the difference in percentage occurrence of breast cancer between woman who are BRCA mutation carriers and woman in the general population by the age of 50.

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[Total: 10]

5 In a dihybrid inheritance, gene **B/b** codes for flower colour while gene **H/h** codes for leaf shape of a plant.

The F1 progeny of a pure-bred plant with red flowers and oval leaves, and another pure-bred plant with yellow flowers and fan-shaped leaves, have red flowers and fan-shaped leaves.

F1 plants then undergo a test cross.

(a) Predict the expected phenotypic ratio in the offspring progeny.

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(b) Identify and explain the dominant traits in this dihybrid cross.

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(c) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratios for the offspring of the test cross.

[3]

(d) Plants are a good choice of experimental organisms for carrying out such crosses and for performing statistical tests.

Compared to plants, humans are less ideal and it is usually more difficult to arrive at reliable conclusions for observations involving humans. Suggest why.

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[Total: 7]

Section B

Answer EITHER 6 OR 7.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Either

6 **(a)** The expression of genes gives rise to proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of specific shapes of proteins in organisms. [8]

(b) Compare and contrast DNA replication with transcription. [7]

[Total: 15]

Or

7 **(a)** Biochemical reactions in cells often involve cyclic metabolic pathways which occur within compartments inside cells.

Compare the Krebs cycle and Calvin cycle in plant cells **and** suggest why the cellular compartmentalisation in both cycles is so important. [7]

(b) Discuss briefly the role of meiosis in evolution **and** explain, with examples, how environmental factors act as forces of natural selection. [8]

[Total: 15]

END OF PAPER